

Total No. of Questions : 8]

SEAT No. :

PE-2524

[Total No. of Pages : 2

[6583]-50

T.E. (Artificial Intelligence & Data Science)

DATA SCIENCE

(2019 Pattern) (Semester - VI) (317529)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Explain Model Selection in detail. [5]
b) Explain in Detail Data Preparation Phase of Data Analytics life cycle. [5]
c) Explain Data Discovery phase in Data analytics Life cycle with example. [8]

OR

- Q2)** a) Write a short note on i) ETL ii) Common tools for model building. [5]
b) Why communication is important in Data analytics Life cycle. [5]
c) Explain Data Analytics Life Cycle with diagram. [8]

- Q3)** a) Write a short note on Association rule. [5]
b) Describe Regression Analysis with example. [5]
c) Explain how Apriori algorithm works, with a suitable example. [7]

OR

- Q4)** a) Explain the term Support, Confidence & Lift by taking a suitable example. [5]
b) Write a short note on Decision Tree. [5]
c) Explain the Naïve Bayes Classifier with its Advantages & Disadvantages. [7]

- Q5)** a) Write a short note on clustering with its properties & types. [5]
b) Write a short note on social network analysis. [5]
c) With a suitable example explain the steps involved in K-means algorithm. [8]

OR

P.T.O.

- Q6)** a) Explain confusion Matrix with all its key terms. [5]
b) What is Hyperparameter tuning. What are the various tuning techniques. [5]
c) Write a short note on : [8]
i) Time series Analysis
ii) TF-IDF

- Q7)** a) What is mean by Data Visualization and explain its types? [5]
b) What are the challenges in big data visualization. [5]
c) Explain data visualization tool Tableau. [7]

OR

- Q8)** a) Write a short note on Hadoop Distributed File System and Hive. [5]
b) Explain analytical techniques used in Big Data Visualization. [5]
c) Explain MapReduce paradigm with example. [7]

